

Mathematics and Algorithms of the Mirror Toolkit

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Contents

1	Introduction	2
2	Canonical Principal Axes (CPA) Alignment	2
2.1	Inertia Tensor	2
2.2	Eigendecomposition and Principal Axes	2
2.3	Axis Orientation Convention	3
2.4	Geometric Mode (PCA)	3
2.5	Degeneracy Handling	3
3	Relationship to the Kabsch Algorithm	3
4	Sohncke Mirror Transformation	4
4.1	Reflection Matrices	4
4.2	Relation to Chirality	4
4.3	Application to Sohncke Space Groups	4
5	Pore Detection	5
5.1	Grid-based Distance Field	5
5.2	Pore Centre and Radius	5
5.3	Cylindrical Projection	5
5.4	VDW Heatmap	5
6	Morphology Parameters	5
6.1	Circularity Index (CI)	5
6.2	Fourier Power Spectrum	6
6.3	Corrugation Factor (CF)	6
6.4	Mean Curvature (curv_H)	6
6.5	Helicity	6
6.6	Radial Skewness (skew_r)	6
7	Guest Orientation Scanning	7
7.1	Rotation Parametrisation	7
7.2	VDW Clash Detection	7
7.3	Three-Region Partition (Rolling Sphere)	7
8	Space Group Analysis	7
9	Summary of Parameter Meanings	8

1 Introduction

This document provides a detailed mathematical description of the core algorithms implemented in the Mirror (Chiral Toolkit) platform. It covers the Canonical Principal Axes (CPA) alignment, the Kabsch algorithm, Sohncke mirror transformations, pore detection and morphology parameters, and guest-orientation accessibility mapping.

2 Canonical Principal Axes (CPA) Alignment

CPA alignment transforms a molecule into a standard orientation by aligning its principal axes of inertia with the Cartesian axes. This provides a unique, orientation-independent reference frame.

2.1 Inertia Tensor

For a set of N atoms with Cartesian positions $\{\mathbf{r}_i = (x_i, y_i, z_i)\}$ and masses m_i , the inertia tensor $\mathbf{I}$ is a 3×3 symmetric matrix:

$$\mathbf{I} = \begin{pmatrix} I_{xx} & I_{xy} & I_{xz} \\ I_{xy} & I_{yy} & I_{yz} \\ I_{xz} & I_{yz} & I_{zz} \end{pmatrix}, \quad (1)$$

with components:

$$I_{xx} = \sum_i m_i (y_i^2 + z_i^2), \quad (2)$$

$$I_{yy} = \sum_i m_i (x_i^2 + z_i^2), \quad (3)$$

$$I_{zz} = \sum_i m_i (x_i^2 + y_i^2), \quad (4)$$

$$I_{xy} = - \sum_i m_i x_i y_i, \quad (5)$$

$$I_{xz} = - \sum_i m_i x_i z_i, \quad (6)$$

$$I_{yz} = - \sum_i m_i y_i z_i. \quad (7)$$

The coordinates are first centred at the centre of mass:

$$\mathbf{r}'_i = \mathbf{r}_i - \mathbf{r}_{\text{COM}}, \quad \mathbf{r}_{\text{COM}} = \frac{\sum_i m_i \mathbf{r}_i}{\sum_i m_i}. \quad (8)$$

2.2 Eigendecomposition and Principal Axes

The inertia tensor is diagonalised via eigenvalue decomposition:

$$\mathbf{I} \mathbf{v}_k = \lambda_k \mathbf{v}_k, \quad k = 1, 2, 3, \quad (9)$$

where λ_k are the principal moments of inertia (eigenvalues) and $\mathbf{v}_k$ are the principal axes (eigenvectors). By convention, the eigenvalues are sorted in ascending order: $\lambda_1 \leq \lambda_2 \leq \lambda_3$. The eigenvector $\mathbf{v}_1$ (corresponding to the smallest moment) is assigned to the $+x$ axis, $\mathbf{v}_2$ to $+y$, and the third axis is obtained from the cross product $\mathbf{v}_3 = \mathbf{v}_1 \times \mathbf{v}_2$ to ensure a right-handed coordinate system.

2.3 Axis Orientation Convention

The sign of each principal axis is ambiguous. Mirror resolves this by orienting each axis toward the side with greater atomic mass (for mass-weighted mode) or more atoms (for geometric mode):

$$\mathbf{v}_k \leftarrow \begin{cases} +\mathbf{v}_k & \text{if } \sum_{(\mathbf{r}'_i \cdot \mathbf{v}_k) > 0} m_i \geq \sum_{(\mathbf{r}'_i \cdot \mathbf{v}_k) < 0} m_i, \\ -\mathbf{v}_k & \text{otherwise.} \end{cases} \quad (10)$$

2.4 Geometric Mode (PCA)

In geometric mode, the covariance (scatter) matrix is used instead of the inertia tensor:

$$\mathbf{S} = \sum_i \mathbf{r}'_i (\mathbf{r}'_i)^\top, \quad (11)$$

which is equivalent to PCA with equal weights. The eigenvalue decomposition and axis assignment proceed identically, but the degeneracy criteria differ (see below).

2.5 Degeneracy Handling

A molecule is flagged as degenerate when its principal axes are not uniquely defined. The conditions depend on the mode:

- **Inertia tensor (mass mode):** degenerate if $\lambda_2 < \varepsilon$ (0D or 1D molecule). Planar molecules are NOT degenerate because the perpendicular-axis theorem guarantees a non-zero out-of-plane moment.
- **Covariance matrix (geom mode):** degenerate if any eigenvalue is zero (rank-deficient). Planar molecules are degenerate here.

Additionally, if the eigenvalues are nearly equal (spherical or symmetric top), the axes are considered degenerate:

$$\min \left(\frac{|\lambda_3 - \lambda_2|}{\lambda_3}, \frac{|\lambda_2 - \lambda_1|}{\lambda_3}, \frac{|\lambda_3 - \lambda_1|}{\lambda_3} \right) < \varepsilon_{\text{gap}} = 10^{-6}. \quad (12)$$

In degenerate cases, the centred coordinates are returned without rotation, and the rotation matrix is set to the identity.

3 Relationship to the Kabsch Algorithm

The Kabsch algorithm finds the optimal rotation matrix $\mathbf{R}$ that minimises the root-mean-square deviation (RMSD) between two sets of points $\{\mathbf{p}_i\}$ and $\{\mathbf{q}_i\}$:

$$\min_{\mathbf{R}} \sum_i \|\mathbf{R}\mathbf{p}_i - \mathbf{q}_i\|^2. \quad (13)$$

The solution involves the cross-covariance matrix $\mathbf{H} = \mathbf{P}^\top \mathbf{Q}$ and its singular value decomposition (SVD):

$$\mathbf{H} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^\top, \quad \mathbf{R} = \mathbf{V} \mathbf{U}^\top. \quad (14)$$

Relation to CPA. CPA differs from Kabsch in both purpose and method:

- **Purpose:** Kabsch aligns two different conformations of the same molecule. CPA aligns a single molecule to a canonical reference frame independent of any external target.
- **Matrix:** Kabsch uses the cross-covariance between two point sets. CPA uses the inertia tensor (mass-weighted auto-covariance) of a single point set.
- **Output:** Kabsch gives the rotation between two structures. CPA gives the absolute orientation relative to the principal axes.

Unified view. Both algorithms ultimately compute an eigendecomposition (CPA) or SVD (Kabsch) of a 3×3 matrix and extract a rotation. If one performs CPA on a structure and then applies the inverse rotation, the principal axes align with Cartesian axes — analogous to rotating one structure onto another’s principal frame.

4 Sohncke Mirror Transformation

4.1 Reflection Matrices

A mirror operation about a plane through the origin (or the molecular centroid) is represented by a Householder reflection matrix. The three available planes correspond to:

$$\mathbf{M}_{YZ} = \text{diag}(-1, 1, 1) \quad (\text{invert } x), \quad (15)$$

$$\mathbf{M}_{XZ} = \text{diag}(1, -1, 1) \quad (\text{invert } y), \quad (16)$$

$$\mathbf{M}_{XY} = \text{diag}(1, 1, -1) \quad (\text{invert } z). \quad (17)$$

4.2 Relation to Chirality

If a molecule is chiral, its mirror image is a non-superimposable enantiomer. The CPA alignment of the mirrored structure (MCPA) enables direct comparison: for a chiral molecule, MCPA will differ from CPA by more than a simple rotation. The difference in principal moments between CPA and MCPA provides a quantitative measure of molecular chirality.

4.3 Application to Sohncke Space Groups

In crystallography, a Sohncke space group contains only orientation- preserving operations (rotations, translations). Mirror operations are present only in non-Sohncke groups. The Mirror toolkit classifies space groups into three types:

- **sgType 1** (Sohncke, enantiomorphic): space groups that allow chiral crystal structures.
- **sgType 2** (Sohncke, non-enantiomorphic): Sohncke groups without enantiomorphic pairs.
- **sgType 3** (non-Sohncke): contain improper rotations (mirrors, inversions).

The set of Sohncke space groups (65 groups) is hardcoded from the International Tables of Crystallography, and the enantiomorphic subset (11 pairs) is similarly defined.

5 Pore Detection

5.1 Grid-based Distance Field

The pore detection pipeline operates on a supercell of the CIF. Atoms are projected onto the ab -plane, and a dense 2D grid is created:

$$\mathbf{G} = \{(x_i, y_j) \mid x_i = x_{\min} + i\Delta x, y_j = y_{\min} + j\Delta y\}, \quad (18)$$

with resolution n_{grid} (default 200). The distance from each grid point to the nearest atom centre (minus that atom's VDW radius) is computed:

$$d(\mathbf{g}) = \min_k (\|\mathbf{g} - \mathbf{a}_k\| - r_k^{\text{VDW}}), \quad (19)$$

where $\mathbf{a}_k$ and r_k^{VDW} are the position and VDW radius of atom k .

5.2 Pore Centre and Radius

The largest connected region where $d(\mathbf{g}) \geq r_{\text{probe}}$ (default 1.2 Å) is identified via breadth-first search. The pore centre is the grid point within this region with the maximum $d(\mathbf{g})$, and the pore radius R_1 is d at that point.

5.3 Cylindrical Projection

VDW-surface points are generated via Fibonacci sphere sampling (N points per atom). These points are projected onto a cylinder of radius R_1 centred at the pore centre:

$$\theta = \arctan 2(y - y_0, x - x_0), \quad (20)$$

$$z = z_{\text{VDW}}. \quad (21)$$

The result is a θ - z point cloud that is binned into a 2D histogram for visualisation.

5.4 VDW Heatmap

The VDW heatmap is a smooth Gaussian kernel-density estimate:

$$\rho(\theta, z) = \sum_p A_p \exp\left(-\frac{(\theta - \theta_p)^2}{2\sigma_\theta^2} - \frac{(z - z_p)^2}{2\sigma_z^2}\right), \quad (22)$$

where A_p is the alpha weight of point p ($A_p = 1 - r_p/r_{\text{max}}$, with r_p the radial distance of the point from the pore centre), and σ_θ , σ_z are proportional to the VDW radius of the originating atom ($\sigma = K \cdot r^{\text{VDW}}$, $K = 0.12$).

6 Morphology Parameters

6.1 Circularity Index (CI)

The angular wall-distance profile $r(\theta)$ is computed by binning VDW points into 72 angular bins (5°). Within each bin, the mean radial distance is computed. CI is then:

$$\text{CI} = 1 - \frac{\sigma_r}{\mu_r}, \quad (23)$$

where μ_r and σ_r are the mean and standard deviation across bins. $\text{CI} = 1$ indicates a perfectly circular pore; $\text{CI} < 1$ indicates ellipticity or faceting.

6.2 Fourier Power Spectrum

The mean-subtracted profile $r'(\theta) = r(\theta) - \mu_r$ is Fourier-transformed:

$$F_n = \sum_{k=0}^{N-1} r'(\theta_k) e^{-2\pi i n k / N}, \quad P_n = \frac{|F_n|^2}{\sum_{k=0}^{N-1} |F_k|^2}. \quad (24)$$

The pore label is assigned from the dominant frequency $n \geq 2$: L6 ($n = 6$), L4 ($n = 4$), L3 ($n = 3$), L2 ($n = 2$).

6.3 Corrugation Factor (CF)

The approximate VDW surface area is:

$$A_{\text{VDW}} \approx N_{\text{pts}} \cdot R_1 \cdot \Delta\theta \cdot \Delta z, \quad (25)$$

where $\Delta\theta = 2\pi/n_\theta$ and $\Delta z = \Delta z_{\text{total}}/n_z$. The smooth cylinder area is $A_{\text{smooth}} = 2\pi R_1 \Delta z_{\text{total}}$. Then:

$$\text{CF} = \frac{A_{\text{VDW}}}{A_{\text{smooth}}}. \quad (26)$$

6.4 Mean Curvature (curv_H)

The VDW density field $\rho(\theta, z)$ is smoothed with a Gaussian filter ($\sigma = 1.0$), and the Laplacian is approximated via finite differences:

$$\nabla^2 \rho \approx \frac{\partial^2 \rho}{\partial \theta^2} + \frac{\partial^2 \rho}{\partial z^2}, \quad (27)$$

$$\text{curv_H} = \langle |\nabla^2 \rho| \rangle. \quad (28)$$

6.5 Helicity

The angular profiles of adjacent z -layers are cross-correlated to find the angular shift $\Delta\theta$ that maximises their alignment:

$$C_k(\Delta\theta) = \sum_i r_k(\theta_i) \cdot r_{k+1}(\theta_i + \Delta\theta). \quad (29)$$

The helical twist angle is:

$$\alpha = \arctan\left(\frac{\Delta\theta_{\text{max}} \cdot R_1}{\Delta z_{\text{layer}}}\right). \quad (30)$$

6.6 Radial Skewness (skew_r)

$$\text{skew_r} = \frac{\langle (r - \mu_r)^3 \rangle}{\sigma_r^3}. \quad (31)$$

Positive skewness indicates that the pore wall has more outward- protruding features than inward ones.

7 Guest Orientation Scanning

7.1 Rotation Parametrisation

Each guest orientation is parametrised by two spherical angles (θ, ϕ) defining the direction of the molecule’s $+z$ axis. The rotation matrix that maps the $+z$ axis to direction (θ, ϕ) is computed via the Rodrigues rotation formula:

$$\mathbf{R}(\theta, \phi) = \mathbf{I} + \sin \alpha [\mathbf{k}]_{\times} + (1 - \cos \alpha)(\mathbf{k}\mathbf{k}^{\top} - \mathbf{I}), \quad (32)$$

where $\mathbf{k}$ is the unit rotation axis (cross product of source and target directions) and α is the rotation angle.

7.2 VDW Clash Detection

For each orientation, the guest atom positions $\{\mathbf{g}_j\}$ are computed from the rotation matrix and the guest centroid position $\mathbf{p}$:

$$\mathbf{g}_j = \mathbf{R}\mathbf{g}_j^{(0)} + \mathbf{p}. \quad (33)$$

A clash is registered if any guest–host atom pair violates the VDW separation criterion:

$$\|\mathbf{g}_j - \mathbf{h}_i\| < s \cdot (r_j^{\text{VDW}} + r_i^{\text{VDW}}), \quad (34)$$

where s is the clash scale factor (default 0.75). The fraction of clash-free orientations is reported as the accessible fraction.

7.3 Three-Region Partition (Rolling Sphere)

The guest’s minimum enclosing VDW sphere radius R_{env} is computed as:

$$R_{\text{env}} = \max_j (\|\mathbf{g}_j - \mathbf{g}_{\text{COM}}\| + r_j^{\text{VDW}}). \quad (35)$$

A rolling-sphere algorithm then determines, for each position (x, y) in the pore cross-section:

- **Free** (green): the envelope sphere ($R = R_{\text{env}}$) fits without clashes.
- **Restricted** (amber): only the centroid ($R = 0$) fits.
- **Blocked** (red): neither fits.

8 Space Group Analysis

Space-group detection uses pymatgen’s `SpacegroupAnalyzer` at multiple tolerance levels:

$$\text{symprec} \in \{0.01, 0.05, 0.1, 0.2\}. \quad (36)$$

The confidence score is the fraction of tolerances that agree on the assigned group. Sohncke and enantiomorphic classifications are determined by lookup tables compiled from the International Tables.

9 Summary of Parameter Meanings

Parameter	Symbol	Physical meaning
CI	CI	Pore circularity; 1 = perfect circle
CF	CF	Wall corrugation; 1 = rough wall
curv_H	$\langle \nabla^2 \rho \rangle$	Mean wall curvature
power_2	P_2	2-fold Fourier power (ellipticity)
power_3	P_3	3-fold Fourier power (trigonal)
power_4	P_4	4-fold Fourier power (square)
power_6	P_6	6-fold Fourier power (hexagonal)
helicity	α	Helical twist between layers (deg)
skew_r	γ_r	Wall-distance skewness
R_1 / R_p	R_1	Pore radius (Å)
R_e	R_e	Expand radius (Å)
PLD	d_{PLD}	Pore-limiting diameter (Å)
LCD	d_{LCD}	Largest cavity diameter (Å)
I_1/I_3	I_1/I_3	Principal-moment ratio
θ_{steps}	N_θ	Polar sampling for orientation scan
ϕ_{steps}	N_ϕ	Azimuthal sampling for orientation scan
s_{clash}	s	VDW clash scale factor
r_{probe}	r_p	Probe radius for pore detection (Å)
n_{grid}	N_{grid}	Grid resolution for distance field
K	K	Gaussian sigma scaling factor (VDW heatmap)